

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Efficacy and Safety of KarXT for the Treatment of Manic Episodes in Bipolar-I Disorder
Short Title/Acronym: BALSAM-1 **Protocol Number:** CN012-0036 **NCT Number:** NCT06951698 **Sponsor Name:** Bristol-Myers Squibb
Investigational Product: KarXT (xanomeline-trospium / trospium chloride and xanomeline tartrate) **Phase of Development:** Phase 3
Medical Monitor: Deepak L. Bhatt, MD (Chief Medical Officer, Bristol-Myers Squibb)

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the efficacy of KarXT compared to placebo in treating symptoms of mania during a 3-week inpatient period. **Secondary Objectives:** To evaluate the efficacy of KarXT on Clinical Global Impressions-Bipolar (CGI-BP) scale at Week 3, and to assess overall safety and tolerability.

2.2 Background & Rationale To address the unmet medical need for novel, well-tolerated treatments for manic episodes in Bipolar-I Disorder. Unlike standard treatments that mainly rely on mood stabilizers and dopamine-focused antipsychotic medications, KarXT offers a fresh approach. By combining xanomeline (a muscarinic receptor agonist) with trospium (a peripheral muscarinic antagonist), the therapy targets the brain's cholinergic system to reduce manic symptoms while potentially limiting peripheral side effects.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration **Target N:** 274 participants
Number of Sites: 34 locations globally **Estimated Study Start:** 2024 **Estimated Primary Completion:** Projected data readout 2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Ages 18-65; All Genders. Primary diagnosis of Bipolar-I disorder established by a comprehensive psychiatric evaluation. 2. Experiencing an acute episode or relapse of mania or mania with mixed features (≤ 3 weeks)

requiring hospitalization. 3. Young Mania Rating Scale (YMRS) score of ≥ 20 and Clinical Global Impressions-Bipolar (CGI-BP) ≥ 4 at Screening and at Baseline.

4.2 Exclusion Criteria 1. Any primary Diagnostic and Statistical Manual of Mental Disorders (DSM-5-TR) disorder other than Bipolar-I (except mild anxiety). 2. History of irritable bowel syndrome (with or without constipation) or serious constipation requiring treatment within the last 6 months. 3. History or high risk of urinary retention, gastric retention, untreated narrow-angle glaucoma, rapid cycling bipolar disorder, or high suicide risk.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: KarXT capsules administered twice daily (BID), based on tolerability. **Route of Administration:** Oral **Duration of Treatment:** 3-week inpatient treatment period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard medical care that does not conflict with the trial protocol. **Prohibited:** All psychotropic medications (must be washed out in no more than 14 days prior to the first dose of the study drug).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Psychiatric evaluation, YMRS & CGI-BP scaling, Psychotropic medication washout
Baseline	Day 0	Randomization, Inpatient admission, First Dose
Interim	Weeks 1-2	Continuous inpatient monitoring, safety tracking, symptom evaluation
End of Study	Week 3	Final YMRS & CGI-BP assessment, transition to safety follow-up (up to Week 7)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline in Young Mania Rating Scale (YMRS) at Week 3. **Measurement Method:** Clinician-administered YMRS assessment tool.

7.2 Secondary & Exploratory Endpoints * Change from baseline in Clinical Global Impressions-Bipolar (CGI-BP) at Week 3. * Overall safety and tolerability assessment profile.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous inpatient observation to track events consistent with xanomeline-trospium activity at muscarinic receptors (e.g., GI events, nausea, constipation, urinary retention). **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal safety committee managed by Bristol-Myers Squibb.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N=274$ appropriately powered to detect significant differences in YMRS scores between the active and placebo arms. **Handling of Missing Data:** Standard statistical imputation methods for psychiatric scales (e.g., Mixed-Effect Model Repeated Measure [MMRM]).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous onsite monitoring during the 3-week inpatient period. **Audit Trail:** eCRF locking, tracking of rater concordance, and strict study drug accountability.

11. Study Identifiers & Governance

Primary ID: CN012-0036 **Registry Number:** NCT06951698 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** BALSAM-1 **Official Regulatory Title:** A Study to Evaluate the Efficacy and Safety of KarXT for the Treatment of Manic Episodes in Bipolar-I Disorder

12. Clinical Framework (Bipolar-I Specific)

Primary Condition: Bipolar-I Disorder (Manic Episodes) **Diagnosis Threshold:** YMRS score ≥ 20 and CGI-BP ≥ 4 **Medical Methodology:** Cholinergic system targeting (Muscarinic receptor agonist/antagonist combination) **Optimization Target:** Reduction in manic symptoms and overall YMRS total score

13. Study Procedures & Methodology

Management Pathway: Inpatient psychiatric hospitalization and observation **Education Protocol (HCP):** YMRS and CGI-BP

administration training for scale raters **Education Protocol (Patient):** Informed consent and inpatient study procedure orientation **Quality Audit Mechanism:** Standard clinical trial oversight for scale rater concordance and protocol adherence

14. Observation & Follow-Up Schedule

Total Duration: Up to 7 weeks (including screening, treatment, and follow-up) **Onsite Visit Cadence:** Continuous 3-week inpatient stay **Remote Monitoring Frequency:** N/A (Inpatient setting) **Checklist Review Interval:** Daily to weekly clinical assessments during the 3-week hospitalization

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					